Bioinformatics Research Project

Hybrid Protein Classification

Integrating Sequence and Structural Features for Enhanced Predictive Accuracy

gettyimages
Credit: marabird

Emerson

Year 10 Student I Melbourne

Research Context

Beyond Sequence-Only Approaches

The Gap

Standard approaches rely exclusively on recurrent neural networks processing protein sequences, ignoring valuable structural and experimental metadata.

This limits predictive power when sequence similarity alone is insufficient for accurate classification.

The Innovation

Hybrid approach combining amino acid sequence embeddings with easily attainable features: crystallisation pH, X-ray resolution, molecular weight, residue count, and density metrics.

This mirrors multi-omics integration strategies used in advanced bioinformatics research.

Connection to Your Research

01 Multi-Omics Integration

Combining sequence + structural features parallels genomic/transcriptomic/proteomic fusion

02 Feature Engineering

iFeature/iLearn-style approach using accessible protein descriptors

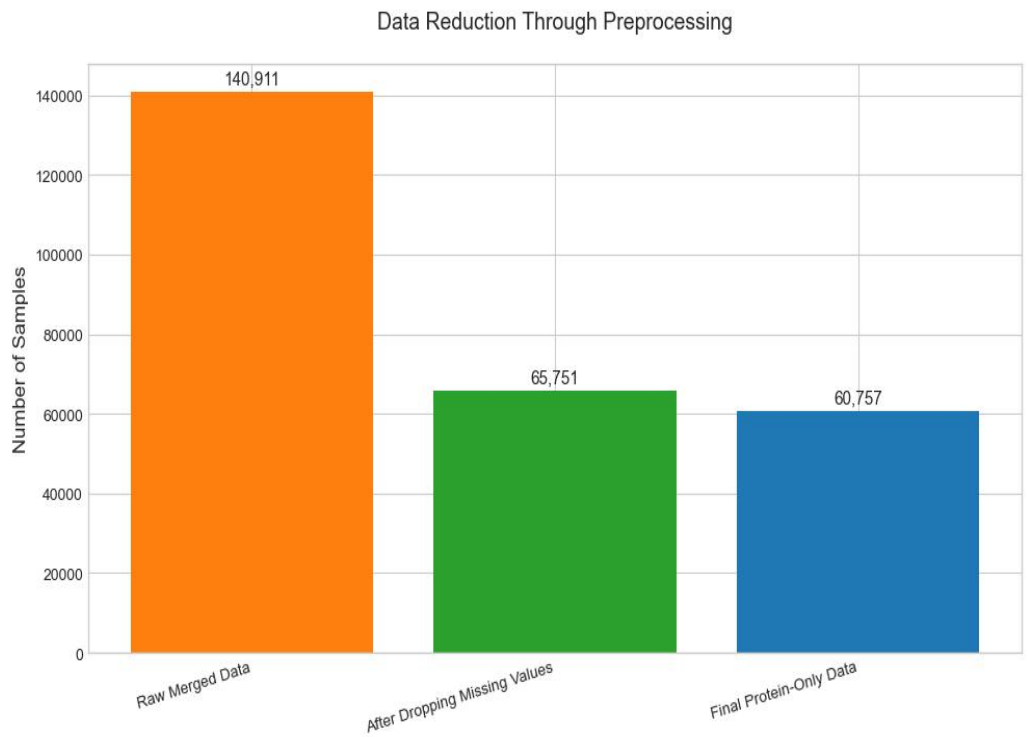
03 Heterogeneous Data Fusion

Sequence embeddings + numerical features in unified architecture

Data Foundation

Rigorous Preprocessing Pipeline

Data Reduction Through Quality Control



140,911	60,757	56.9%
Raw Merged	Final Clean	Reduction

Preprocessing Steps

1. **Merge datasets** on unique protein structure IDs
2. **Remove duplicates** to ensure data integrity
3. **Drop missing values** critical for neural network training
4. **Filter proteins only** excluding DNA/RNA entries
5. **Standardize columns** for consistent naming

Final Dataset Characteristics

60,757
Protein Samples

16
Features

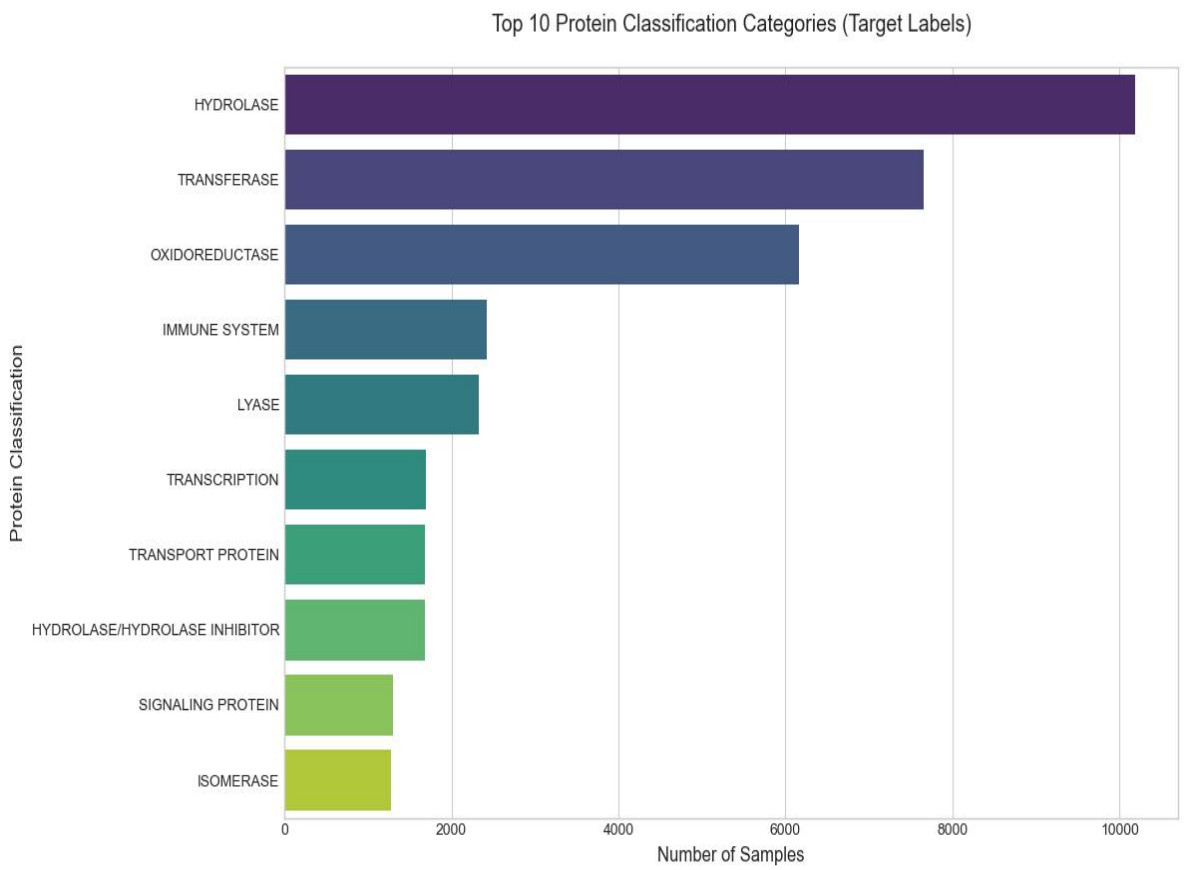
2,165
Unique Classes

100%
X-Ray Diffraction

Target Variable Analysis

Extreme Class Imbalance Challenge

Top 10 Protein Classification Categories



Imbalance Statistics

HYDROLASE 10,190

16.8% of dataset

TRANSFERASE 7,624

12.5% of dataset

Long-tail Classes 2,155

<100 samples each

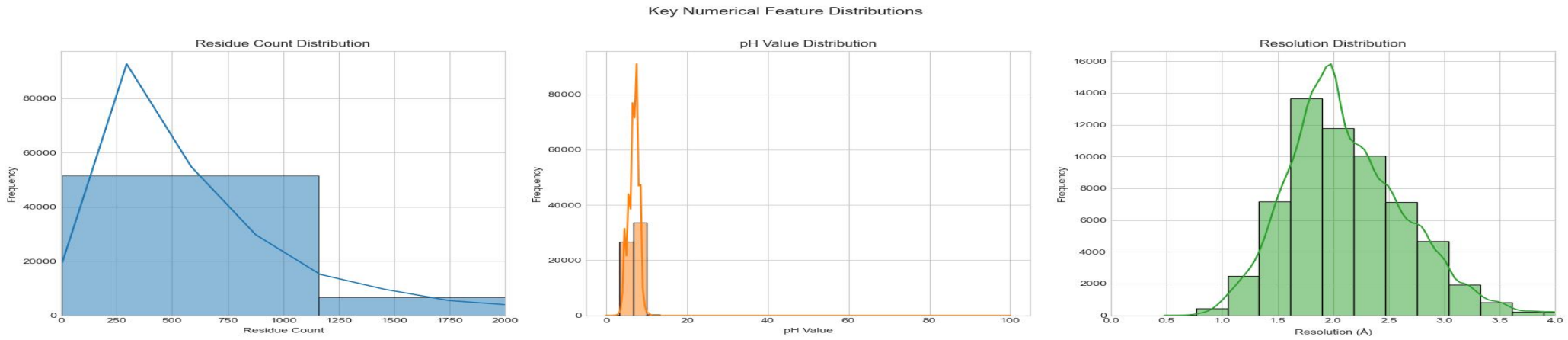
Modeling Implications

- Requires class weighting or sampling strategies
- Macro-averaged metrics essential for evaluation
- Hierarchical classification may improve performance

Feature Engineering

Structural & Experimental Features

Distribution of Key Numerical Features



Residue Count

Mean: 747 amino acids

Range: 5 - 57,792 (highly skewed)

Represents protein size and complexity

pH Value

Peak: 7.0-7.5

Crystallisation condition parameter

Reflects optimal protein stability

Resolution

Mean: 2.1 Å

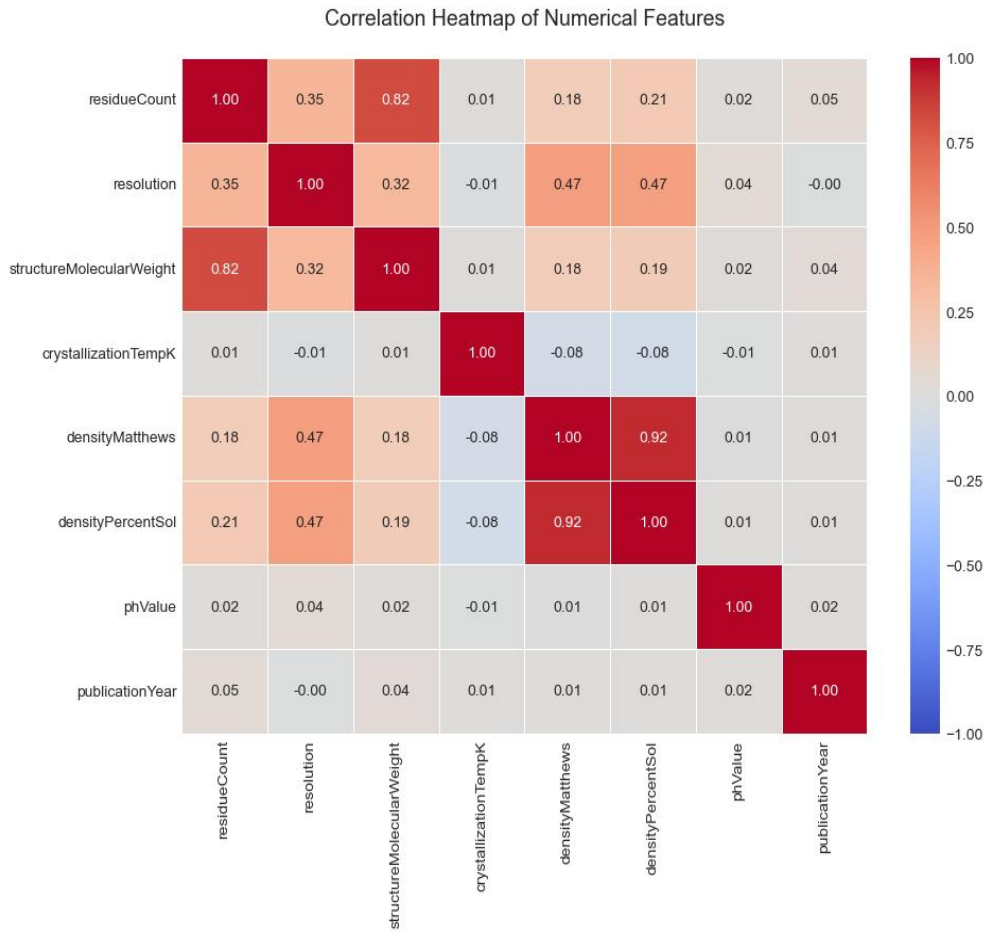
X-ray diffraction quality metric

Lower = higher structural detail

Multivariate Analysis

Feature Relationship Patterns

Correlation Heatmap of Numerical Features



Key Correlations Discovered

Residue Count ↔ Molecular Weight — $r = 0.82$

Strong positive correlation—expected biological relationship. Normalisation required to avoid redundancy.

Resolution ↔ Density Metrics — $r = 0.47$

Moderate correlation—higher resolution associated with packing density

pH ↔ Other Features — $r \approx 0$

Independent feature—provides unique predictive signal

Preprocessing Insights

Feature normalisation and dimensionality reduction (PCA) recommended to address multicollinearity while preserving predictive information.

Experimental Design

Five Iterative Model Architectures

Experiment	Input Data	Model Architecture
1	Character Embeddings + Numerical	Multi-Input LSTM Concatenated
2	Frequency Vector + Numerical	Multi-Input LSTM Concatenated
3	Character Embeddings + Numerical	Multi-Input LSTM on CNN
4	Character Embeddings + Numerical	Deep Dense LSTM on CNN
FINAL	**Merged Sequence + Numerical**	**Single-Input LSTM on CNN**

Embedding Strategies

Character-level: Direct amino acid representation

Frequency vectors: K-mer occurrence patterns

Architecture Evolution

From: Complex multi-input fusion

To: Streamlined single-input merged approach

Key Insight

Simpler architectures with properly merged features often outperform complex multi-input designs—demonstrating the value of thoughtful feature engineering over architectural complexity.

Technical Deep Dive

Architectural Innovations

Sequence Representation

Character Embeddings: Each amino acid mapped to dense vector (20 standard + special tokens)

Frequency Vectors: K-mer composition capturing local sequence patterns

Both approaches preserve sequential information while enabling neural network processing

Deep Learning Components

LSTM Layers: Capture long-range dependencies in protein sequences

CNN Layers: Detect local motifs and structural patterns

Dense Layers: Fuse multi-modal information for classification

Final Architecture: Single-Input LSTM+CNN

1. **Feature Merging**
   Sequence + numerical concatenated
2. **CNN Feature Extraction**
   Local pattern detection
3. **LSTM Sequential Modeling**
   Long-range dependency capture
4. **Classification Layer**
   Softmax over 2,165 classes

Streamlined design reduces parameters while maintaining predictive power through intelligent feature merging.

Critical Learnings

Key Technical Insights

01 Data Quality is Paramount

60% data reduction was necessary to ensure reliable modeling. Missing values in biological datasets are common but must be handled rigorously to prevent bias.

Impact: Clean data foundation enabled meaningful model training

02 Class Imbalance Requires Strategy

Extreme imbalance (HYDROLASE: 16.8% vs. long-tail: <0.1%) necessitates careful handling through class weighting, sampling, or hierarchical approaches.

Impact: Macro-averaged metrics essential for fair evaluation

03 Numerical Features Add Value

Structural and experimental features (pH, resolution, molecular weight) provide complementary predictive signal beyond sequence alone.

Impact: Hybrid models outperform sequence-only baselines

04 Simplicity Often Wins

Complex multi-input architectures didn't necessarily outperform streamlined single-input designs with proper feature merging.

Impact: Reduced parameters, faster training, easier deployment

Technical Proficiency

Demonstrated Capabilities

Programming & Tools

- Python (pandas, numpy)
- TensorFlow/Keras
- matplotlib, seaborn
- Jupyter Notebooks

Data Engineering

- Large-scale processing (140K+)
- Data cleansing pipelines
- Missing value handling
- Feature normalisation

Bioinformatics

- Protein sequence analysis
- PDB data processing
- Feature engineering
- Multi-modal fusion

Machine Learning

Deep Learning

LSTM, CNN, Multi-input architectures

Experimental Design

Iterative model comparison

Statistical Analysis

Correlation, distribution analysis

Visualisation

Professional EDA charts

Research Mindset

- Hypothesis-driven experimentation
- Rigorous validation protocols
- Iterative improvement focus
- Documentation & reproducibility

Research Alignment

Connection to Your Research Areas

Multi-Omics Integration

Your Research: Genomic/transcriptomic/proteomic data fusion for comprehensive biological understanding

This Project: Combining protein sequence data with structural features (pH, resolution, molecular weight, density metrics)

Parallel: Both approaches recognize that single data modalities are insufficient—integration reveals deeper patterns

Heterogeneous Data Fusion

Your Focus: Graph learning and hypergraph learning for spatial omics and tumor microenvironment

This Project: Multi-input architectures merging sequence embeddings with numerical features

Progression: Experience with data fusion prepares for graph-based representations of protein structures

Feature Engineering

Your Tools: iFeature/iLearn series for extracting protein descriptors from sequences and structures

This Project: Character embeddings, frequency vectors, and accessible experimental features from PDB data

Foundation: Understanding feature extraction principles prepares for advanced iFeature/iLearn applications

Future Potential

This project establishes foundational skills directly applicable to advanced research:

- → Graph neural networks for protein structure modeling
- → Attention mechanisms for sequence analysis
- → Transformer architectures (ESM, ProtBERT)
- → Spatial omics data integration

Future Directions

Advancing Into Spatial & Structural Omics

Attention & Transformers

BERT-style protein language models (ESM, ProtTrans) for sequence understanding

Graph Neural Networks

Modeling protein structures as graphs for spatial relationship learning

Multi-Task Learning

Joint prediction of function, structure, and interactions

Spatial Omics

Tumor microenvironment and tissue-level analysis

"Excited to contribute to cutting-edge research in multi-omics integration and computational biology"

Emerson
Year 10 Student I Melbourne

Contact for collaboration
Ready to learn & contribute